ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594
2. Isolating a specific category makes the discovered frequent patterns more domain-relevant and prevents unrelated topics from distorting the support of word co-occurrences.

AI used for this answer: No

If yes, how:

External references (links):

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

Directly dropping 40% of the data is problematic because we would lose a massive amount of still-useful text data, which reduces the effective sample size. Furthermore, if the missing labels occur systematically rather than randomly, the remaining data may no longer be representative of the original corpus, potentially distorting downstream term frequencies, support, and category distributions.

Objective alternative: Train a text classifier using the 60% of documents that currently have a category_name, and then use this model to predict the missing labels for the remaining 40%.

AI used for this answer: No

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No, the proportions do not match perfectly due to sampling variability. For example, comp.graphics increased from 25.88% to 27.80%, showing that simple random sampling cannot guarantee identical class ratios.

2. On highly imbalanced datasets, the risk is minority class underrepresentation or complete exclusion. A random sample may capture too few minority instances, causing critical information loss and poor model performance on minority classes. Stratified sampling can mitigate this risk by better preserving the original class proportions.

AI used for this answer: Yes

If yes, how: grammatical help

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html (stratify parameter)

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The total number of features (columns) increases from 35,482 to 35,788, as the previously excluded stopwords are reintroduced into the vocabulary.

2. The overall matrix sparsity decreases from 99.6772% to 99.5470%. Because stopwords frequently appear across many documents, adding them back introduces many non-zero entries, increasing the proportion of non-zero elements and thereby reducing the percentage of zeros in the matrix.

Only 306 columns are added, although scikit-learn's English stop list contains 318 words. Two of them, 'a' and 'i', are single letters that the default token pattern (two or more word characters) never produces, and the remaining 10 never occur in these 2,257 documents, so they were never part of the vocabulary.

AI used for this answer: Yes

If yes, how: Review of my draft; AI explained why only 306 instead of 318 stop words are added back, which I added as an extra paragraph; grammar.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.CountVectorizer.html

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Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

1. The vocabulary shrinks sharply as the threshold rises: 35,482 terms are kept at 0.0, 5,089 at 0.01 and 1,491 at 0.05, i.e. only about 14% and 4% of the original vocabulary. At 0.0 nothing is removed, because every term appears in at least one document and therefore never has exactly zero variance.

2. Words with near-zero variance are typically those that appear in very few documents. Rare words that appear only once in a small number of documents often have low variance because their counts are zero in almost all documents.

For a term that appears once in d of the N = 2,257 documents, the variance is roughly $(d/N)(1 - d/N)$. A threshold of 0.01 therefore removes terms found in fewer than about 23 documents, and 0.05 those found in fewer than about 120. Frequent words are not the ones removed here: stop words were already excluded, and common content words have high variance because their counts grow with document length.

3. Removing near-zero variance features is generally considered a useful dimensionality reduction technique because features that barely change across documents often provide limited information for distinguishing between observations. Eliminating these features reduces computational costs, memory usage, and potentially irrelevant information.

However, this approach does not guarantee unchanged model accuracy. Some rare words may have low variance but still be highly informative for identifying specific categories. Therefore, the impact of feature removal should be evaluated using validation data to ensure that important predictive information is preserved.

AI used for this answer: Yes

If yes, how: Review and correction of my own draft (answer 2), concept check of the variance formula, grammar.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.VarianceThreshold.html

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. The top-correlated words are a mixture of domain-specific terminology and general words. In the Spearman results, terms such as "graphics," "image," and "3d" are strongly related to computer graphics. However, the Pearson results also include general words such as "thanks," "looking," and "hi." This shows that statistical correlation does not necessarily imply semantic relevance. Overall, the Spearman results contain more clearly graphics-related terminology in the top ten.

2. Pearson correlation measures the strength and direction of a linear relationship between two variables using their original numerical values. In contrast, Spearman correlation measures the strength and direction of a monotonic relationship using ranked values rather than raw data. Therefore, Spearman can detect strong monotonic relationships even when they are not linear, and it is generally less sensitive to extreme values.

3. A word may have a high Spearman correlation but a lower Pearson correlation when its frequency consistently ranks higher in documents belonging to a specific topic, while a few extreme word counts distort the linear relationship.

For example, suppose the word "rendering" never appears in non-graphics documents but appears 1, 2, 3, and 1,000 times in graphics documents. The extreme count of 1,000 can substantially affect Pearson correlation, while Spearman remains high because the word frequencies in graphics documents consistently rank above those in other categories.

This effect is especially strong in sparse text data. For any given word most documents have a count of 0, so all of them share one low tied rank and every non-zero count is pushed into a few top ranks. The jump from 0 to 1 dominates, which makes Spearman behave almost like a presence/absence measure: a word that appears once in many graphics posts gets a high Spearman value, while its Pearson value can be pulled down by a few documents with very large counts.

AI used for this answer: Yes

If yes, how: Concept explanation of Pearson vs. Spearman on sparse count data (tied ranks), used to extend answer 3; review and grammar.

External references (links):

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.spearmanr.html

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Decreasing minSup from 9 to 3 significantly increases both the execution runtime and the number of discovered patterns. Based on the experimental results, the number of patterns increased from 710 to 9,547, while the runtime increased from 0.1304 to 2.4125 seconds.

This happens because a lower minimum support threshold allows more infrequent itemsets to qualify as frequent patterns, expanding the search space and increasing computational complexity.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

The parameter k in Top-K FAE specifies the number of most frequent patterns to retain, regardless of a predefined minimum support threshold. For example, setting k = 100 or k = 500 produces 100 or 500 patterns, respectively.

In contrast, minSup in MaxFP-Growth specifies the minimum occurrence frequency required for a pattern to be considered frequent. The number of resulting patterns is therefore determined by the dataset and the support threshold rather than a fixed output size.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns are more concise because they include only frequent itemsets that are not subsets of any larger frequent itemset.

Unlike standard frequent pattern mining, which returns all itemsets satisfying the minimum support threshold, maximal frequent pattern mining eliminates redundant subsets. This significantly reduces the number of reported patterns while preserving the largest frequent combinations.

However, this approach sacrifices the individual support information of the omitted subsets.

	Algorithm	Parameter	Runtime	Pattern Count
0	Top-K FAE	100	0.052700	100
1	Top-K FAE	500	0.189247	500
2	MaxFP-Growth	9	0.130436	710
3	MaxFP-Growth	3	2.412459	9547

AI used for this answer: Yes

If yes, how: Review of my draft for correctness and grammar.

External references (links): https://github.com/UdayLab/PAMI

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. The original X_counts matrix has a shape of (2257, 35482), meaning that each document is represented by 35,482 features. However, a standard 2D scatter plot can only display two coordinates per data point.

Therefore, it is not possible to directly visualize all 35,482 dimensions simultaneously in a 2D scatter plot. Although we could select two individual features, doing so would discard most of the information contained in the original feature space.

Dimensionality reduction is necessary to project these high-dimensional documents into a lower-dimensional space while attempting to preserve important relationships between them.

2. Dimensionality reduction algorithms must be applied first. In the helper file visualization_helpers.py ("Plotting helpers for PAMI stats and dimensionality reduction"), compute_dimred_2d_for_filters projects each filtered document-term matrix to two dimensions with PCA (linear) and with t-SNE and UMAP (non-linear manifold learning). Only these 2D/3D coordinates are then drawn with plot_scatter_2d.

AI used for this answer: Yes

If yes, how: Review: AI pointed out that answer 2 was an accidental copy of answer 1 and helped me check which helper functions perform the reduction.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html ; https://umap-learn.readthedocs.io/en/latest/

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?

2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. Based on the 3D visualizations, UMAP shows the clearest category separation among the three dimensionality reduction methods. Some categories form relatively distinct regions or small clusters, although considerable overlap remains.

In contrast, PCA shows heavy overlap around the center, with most data points concentrated in a small region and several outliers extending far from the main cluster.

t-SNE also produces a concentrated distribution, but the four categories remain largely mixed together. Therefore, UMAP provides the most visible category separation in this experiment, although none of the methods produces four completely distinct cluster islands.

2. PCA is a linear dimensionality reduction method that identifies directions of maximum variance rather than maximizing category separation. In high-dimensional text data, large variations in word frequencies or document lengths may dominate the principal components, even when these variations are not useful for distinguishing document categories.

Furthermore, PCA may fail to capture complex non-linear relationships between documents. In contrast, t-SNE and UMAP can preserve non-linear neighborhood structures, potentially revealing clearer category clusters in lower-dimensional space.

3. t-SNE primarily preserves local neighborhood relationships rather than global distances between clusters. As a result, the physical distances between clusters in a t-SNE visualization do not necessarily reflect their actual distances or similarities in the original high-dimensional space.

For example, two clusters appearing close together in a t-SNE plot are not necessarily more similar than two clusters positioned farther apart. Therefore, analysts should focus on local neighborhood structures and avoid interpreting inter-cluster distances as precise measures of similarity.

AI used for this answer: Yes

If yes, how: Review of my draft and a small grammar/spacing fix.

External references (links): https://distill.pub/2016/misread-tsne/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. Four binary columns are created, one for each category. Each category is represented by a unique one-hot vector, such as [1, 0, 0, 0], where only one position is 1 and the remaining positions are 0.

2. Using raw integer labels (0, 1, 2, 3) as numerical input features introduces a false assumption that the categories have a meaningful numerical order and distance.

For example, a distance-based algorithm such as KNN would treat category 0 as closer to category 1 than to category 3, even though these numbers are merely identifiers and do not represent actual relationships between categories.

3. One-hot binarization represents each category as a separate binary feature, where 1 indicates membership in that category and 0 indicates otherwise.

This removes the artificial numerical ordering imposed by integer labels. Under Euclidean distance, all distinct one-hot encoded categories are equally distant from one another, preventing the algorithm from assuming false ordinal relationships.

For example, the Euclidean distance between any two different one-hot vectors, such as [1, 0, 0, 0] and [0, 0, 0, 1], is always $\sqrt{2}$, so no category is closer to one category than to another.

AI used for this answer: Yes

If yes, how: Review of my draft; added the $\sqrt{2}$ distance example to answer 3.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.LabelBinarizer.html

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